



Model Order Reduction in Finite Element Analysis of Phased Array Antennas

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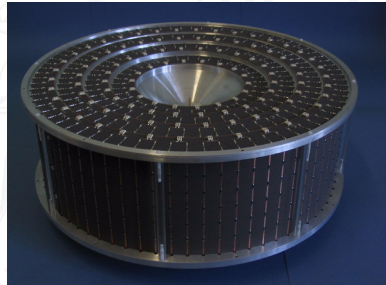
Phased arrays for radar and communication applications



RAT 31 DL - SELEX Sistemi Integrati.
D-band solid state phased array, 3D
surveillance radar



Thomson Master-T Radar - ©Thomson
CSF Air Command and Control Systems



P. Ratajczak, P. Brachat, J.M. Fargeas, "Adaptive
Beam Steering Base Station Antenna: Performances
Optimization of the Reconfigurable EBG Material" -
France Telecom

Numerical analysis of phased arrays: requirements

- ▶ Flexibility
 - ▶ Arbitrary shape of radiating elements
 - ▶ Arbitrary material properties
- ▶ High accuracy and reliability
 - ▶ Full-wave solution of Maxwell's equations



Finite Elements discretization leading to
a linear system of equations

- ▶ Electrically large structures $\Rightarrow 10^5 - 10^8$ unknowns !!!
- ▶ Solutions have to be computed for each scan angle !!!

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Radiation model and Model Order Reduction

- Once reduced, the **radiation model** must preserve the beam steering information

Model parameterization:

The model is expressed in terms of the scan angle (array excitation) and the look angle (radiation pattern)

Finite elements formulation: near fields

Solution of the wave equation for electric field:

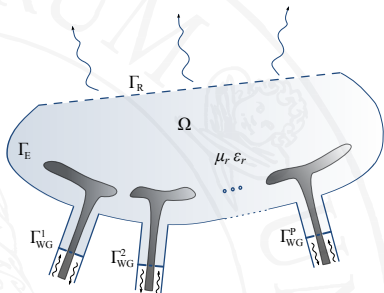
$$\nabla \times \frac{1}{\mu_r} \nabla \times \mathbf{E} - k_0^2 \epsilon_r \mathbf{E} = 0$$

$$k_0 = \omega \sqrt{\epsilon_0 \mu_0}, \quad \zeta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

nel dominio $\Omega \subset \mathbb{R}^3$ con contorno

$$\partial\Omega = \Gamma = \Gamma_E \cup \Gamma_H \cup \Gamma_{WG} \cup \Gamma_R$$

- ▶ $\Omega_h = \bigcup_{m=1}^M \Omega_m \subset \Omega, \quad \Omega_m \cap \Omega_n = \{\emptyset\} \quad \forall m \neq n$
- ▶ $\tilde{\mathbf{E}} := \sum_{j=1}^N \mathbf{e}_j w_j, \quad w_j \in \mathcal{W} \subset \mathcal{H}(\text{curl}; \Omega)$
- ▶ $\int_{\Omega_h} w_i \cdot \left(\nabla \times \frac{1}{\mu_r} \nabla \times \tilde{\mathbf{E}} - k_0^2 \epsilon_r \tilde{\mathbf{E}} \right) d\Omega = 0$



$$\mathbf{A} \mathbf{e}_p = \mathbf{b}_p, \quad 1 \leq p \leq P$$

$$\mathbf{A} \in \mathbb{C}^{N \times N}, \quad \mathbf{b}_p, \mathbf{e}_p \in \mathbb{C}^{N \times 1}$$

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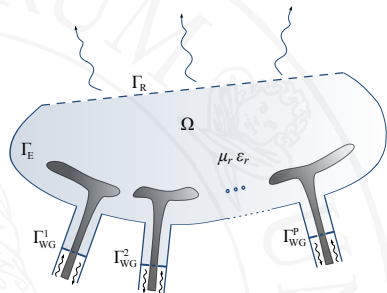
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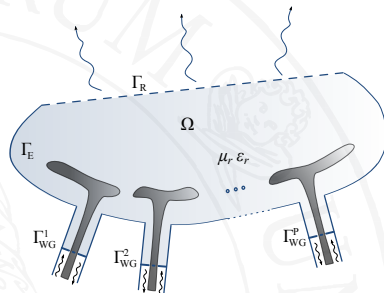
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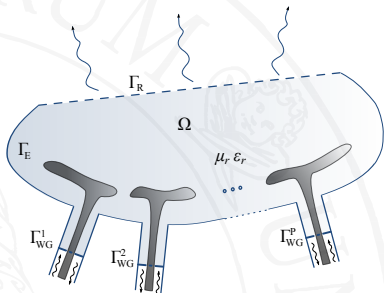
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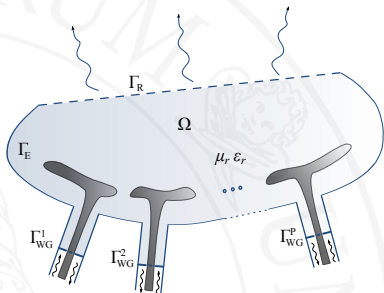
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$$[e_1, \dots, e_N]^T = \mathbf{e} = \sum_{p=1}^P \mathbf{e}_p \quad (\text{expansion coefficients for } \tilde{\mathbf{E}})$$

- ▶ $\nabla \times \mathbf{E} = -j\omega \mathbf{B} \Rightarrow \tilde{\mathbf{H}} = -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}$
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Near field to far field transformation

- **Huygens' principle:** far field derived from near field

$$\tilde{\mathbf{E}} = \tilde{\mathbf{E}}(\mathbf{r}') \text{ e } \tilde{\mathbf{H}} = \tilde{\mathbf{H}}(\mathbf{r}'), \mathbf{r}' \in \Gamma_R$$

$$\check{\mathbf{E}}(\theta, \phi) = j \frac{\zeta_0 k_0}{4\pi} \iint_{\Gamma_R} e^{jk_0 \hat{\mathbf{r}} \cdot \mathbf{r}'} \left[-(\hat{\mathbf{n}} \times \tilde{\mathbf{H}}) + ((\hat{\mathbf{n}} \times \tilde{\mathbf{H}}) \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} + \frac{1}{\zeta_0} (\hat{\mathbf{n}} \times \tilde{\mathbf{E}}) \times \hat{\mathbf{r}} \right] d\Gamma$$

$\hat{\mathbf{r}} = \hat{\mathbf{r}}(\theta, \phi)$ unit vector in the **look angle direction**

$\hat{\mathbf{n}}$ normal on Γ_R outwardly directed from Ω_h

- Radiated field approximation: $\check{\mathbf{H}}(\theta, \phi) = \frac{1}{\zeta_0} \hat{\mathbf{r}} \times \check{\mathbf{E}}(\theta, \phi)$

- Radiation pattern:

$$G_{\hat{\theta}}(\theta, \phi) = 4\pi \frac{|\hat{\theta} \cdot \check{\mathbf{E}}(\theta, \phi)|^2}{2\zeta_0 P_{\text{acc}}}, \quad G_{\hat{\phi}}(\theta, \phi) = 4\pi \frac{|\hat{\phi} \cdot \check{\mathbf{E}}(\theta, \phi)|^2}{2\zeta_0 P_{\text{acc}}}$$

$$G_{\text{tot}}(\theta, \phi) = G_{\hat{\theta}}(\theta, \phi) + G_{\hat{\phi}}(\theta, \phi)$$

$P_{\text{acc}} = P_{\text{inc}} - P_{\text{refl}}$: Accepted power at the feeding ports

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Radiation model: Input-State relation

- $\mathbf{A} \mathbf{e}_p = \mathbf{b}_p \times$ **excitation phasors** for a constructive interference in the scan direction (θ_s, ϕ_s) :

$$\xi_p(\theta_s, \phi_s) = e^{-jk_0(x_p \sin(\theta_s) \cos(\phi_s) + y_p \sin(\theta_s) \sin(\phi_s) + z_p \cos(\theta_s))}$$

(x_p, y_p, z_p) position of p^{th} antenna

- Input-State relation becomes parameterized in the scan direction (θ_s, ϕ_s)

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- Integral relation between near field and radiated electric field
 $\Rightarrow$ **operators on the near field** leading to radiated field in $\hat{\theta}$
 and $\hat{\phi}$ polarizations

$$\check{\mathbf{E}}_{\hat{\theta}}(\theta, \phi) = \hat{\theta} \cdot \check{\mathbf{E}}(\theta, \phi) = \mathcal{L}_{\hat{\theta}}(\check{\mathbf{E}}, \check{\mathbf{H}}; \theta, \phi)$$

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- Operators expressed in 2D Fourier series

$$\mathcal{L}_{\{\hat{\theta}, \hat{\phi}\}}(\check{\mathbf{E}}, \check{\mathbf{H}}; \theta, \phi) = \sum_{q_{\theta}=-\infty}^{\infty} \sum_{q_{\phi}=-\infty}^{\infty} k_{q_{\theta}q_{\phi}\{\hat{\theta}, \hat{\phi}\}}(\check{\mathbf{E}}, \check{\mathbf{H}}) \eta_{q_{\theta}q_{\phi}}(\theta, \phi)$$

$$\eta_{q_{\theta}q_{\phi}}(\theta, \phi) = e^{j(q_{\theta}\theta + q_{\phi}\phi)}, \quad q_{\theta}, q_{\phi} \in \mathbb{Z}$$

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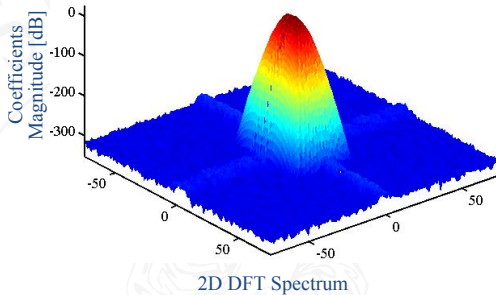
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- Narrow bandwidth operators $\Rightarrow$ 2D-DFT truncation:

$$\mathcal{L}_{\{\hat{\theta}, \hat{\phi}\}}(\tilde{\mathbf{E}}, \tilde{\mathbf{H}}; \theta, \phi) \approx \tilde{\mathcal{L}}_{\{\hat{\theta}, \hat{\phi}\}}(\tilde{\mathbf{E}}, \tilde{\mathbf{H}}; \theta, \phi) = \sum_{q_{\theta}=-Q_{\theta}}^{Q_{\theta}} \sum_{q_{\phi}=-Q_{\phi}}^{Q_{\phi}} k_{q_{\theta} q_{\phi} \{\hat{\theta}, \hat{\phi}\}}(\tilde{\mathbf{E}}, \tilde{\mathbf{H}}) \eta_{q_{\theta} q_{\phi}}(\theta, \phi)$$

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$\Downarrow$

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► State-Output relation parameterized in the look direction
(θ, ϕ)

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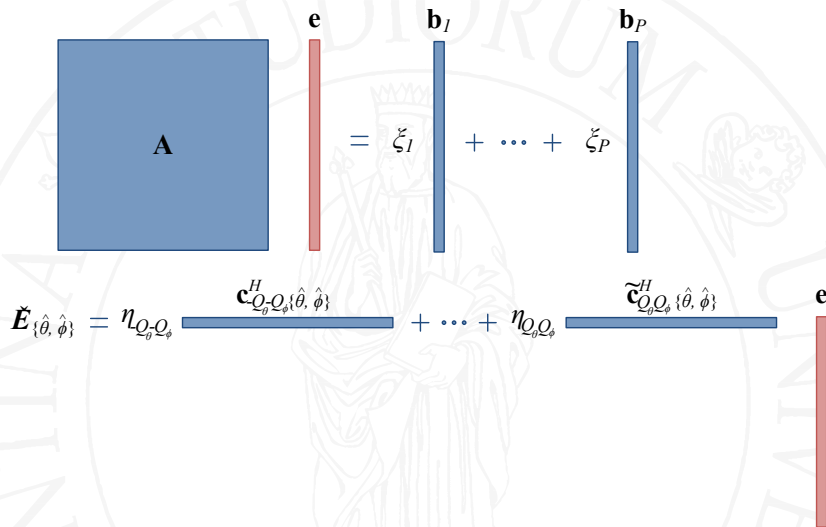
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- State-Output relation parameterized in the look direction (θ, ϕ)

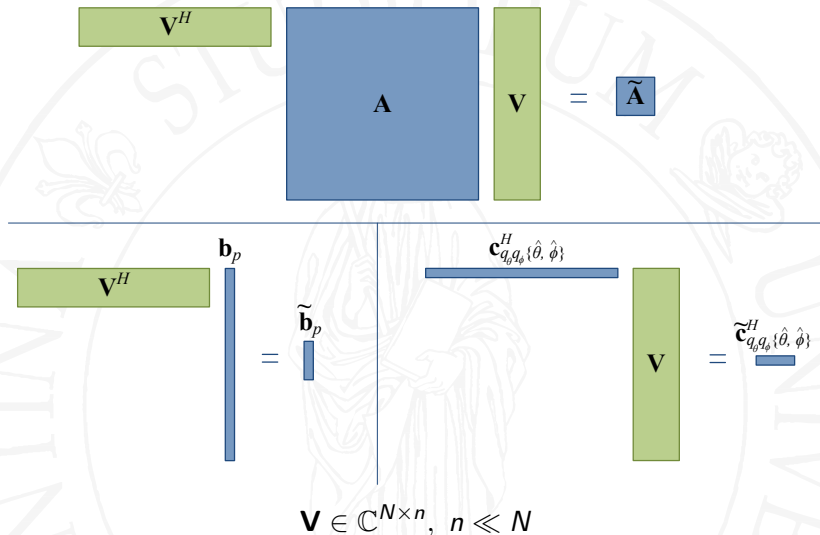
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Full Input-State-Output system



Full Input-State-Output system: projection



Reduced Input-State-Output system

$$\begin{aligned} \tilde{\mathbf{A}} \begin{bmatrix} \tilde{\mathbf{e}} \end{bmatrix} &= \zeta_l \begin{bmatrix} \tilde{\mathbf{b}}_l \end{bmatrix} + \dots + \zeta_P \begin{bmatrix} \tilde{\mathbf{b}}_P \end{bmatrix} \\ \dot{\mathbf{E}}_{\{\hat{\theta}, \hat{\phi}\}} &= \eta_{Q_0 Q_\phi} \frac{\tilde{\mathbf{c}}_{Q_0 Q_\phi \{\hat{\theta}, \hat{\phi}\}}^H}{\text{---}} + \dots + \eta_{Q_0 Q_\phi} \frac{\tilde{\mathbf{c}}_{Q_0 Q_\phi \{\hat{\theta}, \hat{\phi}\}}^H}{\text{---}} \begin{bmatrix} \tilde{\mathbf{e}} \end{bmatrix} \end{aligned}$$

Projection space

- ▶ $\mathbf{V}$ constitutes an **expansion basis** for the solution vector $\tilde{\mathbf{e}}(\theta_s, \phi_s)$: $\mathbf{e}(\theta_s, \phi_s) \approx \mathbf{V}\tilde{\mathbf{e}}(\theta_s, \phi_s) \Rightarrow \mathbf{V}$ must keep information on the near field variations due to beam steering
- ▶ Collection of α solutions linearly independent for the scan directions $(\theta_{s_i}, \phi_{s_i})$, $1 \leq i \leq \alpha \Rightarrow$ orthonormalization $\Rightarrow \mathbf{V}$
- ▶ The model order reduction error is computed as relative error induced to the pattern in $\mathcal{L}^2$:

$$\varepsilon = \frac{\|\mathbf{G}_{tot_{full}} - \mathbf{G}_{tot_{reduced}}\|_2}{\|\mathbf{G}_{tot_{full}}\|_2}$$

Projection space

- ▶ $\mathbf{V}$ constitutes an **expansion basis** for the solution vector $\tilde{\mathbf{e}}(\theta_s, \phi_s)$: $\mathbf{e}(\theta_s, \phi_s) \approx \mathbf{V}\tilde{\mathbf{e}}(\theta_s, \phi_s) \Rightarrow \mathbf{V}$ must keep information on the near field variations due to beam steering
- ▶ Collection of α solutions linearly independent for the scan directions $(\theta_{s_i}, \phi_{s_i})$, $1 \leq i \leq \alpha \Rightarrow$ **orthonormalization** $\Rightarrow \mathbf{V}$
- ▶ The model order reduction error is computed as relative error induced to the pattern in $\mathcal{L}^2$:

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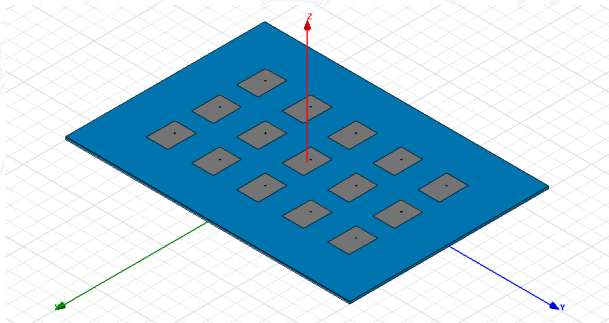
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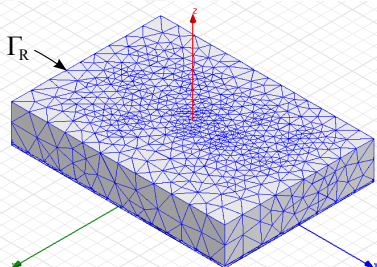
HFSS design of a 3x5 patch antennas array

$$f = 2.35 \text{ GHz}, \quad \Delta x = \Delta y = \lambda_0/2$$



HFSS design of a 3x5 patch antennas array

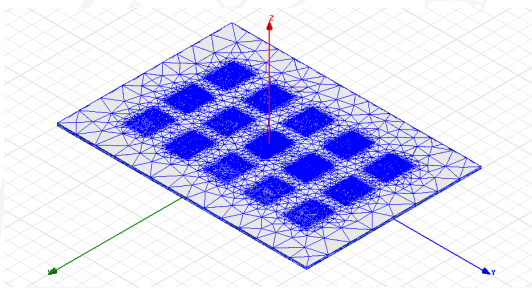
$$f = 2.35 \text{ GHz}, \quad \Delta x = \Delta y = \lambda_0/2$$



- ▶ First order Nedelec basis functions
- ▶ h -adaptive tetrahedral mesh ($h_{\max} \approx \lambda/10$)

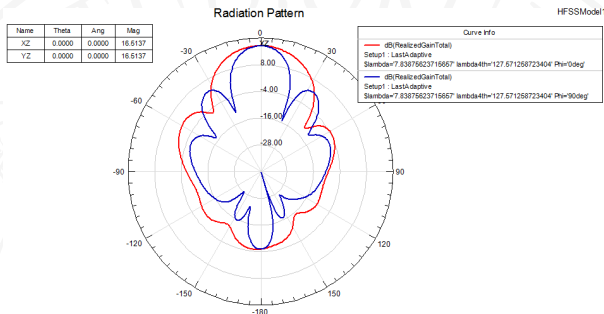
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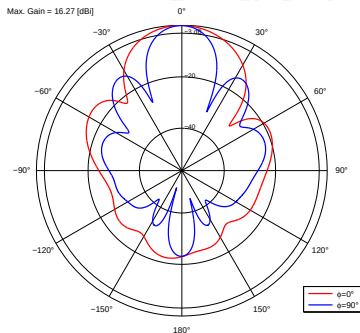
- ▶ First order Nedelec basis functions
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Broadside pattern ($\theta_s=0^\circ, \phi_s=0^\circ$): HFSS



- ▶ 358' 648 DoFs
- ▶ Memory requirements: ≈ 1.5 GB
- ▶ Computation time ≈ 38 min
- ▶ Pattern computation on $\phi = 0^\circ$ and $\phi = 90^\circ$ cut-planes demands ≈ 35 s

Broadside pattern ($\theta_s=0^\circ, \phi_s=0^\circ$): Present code



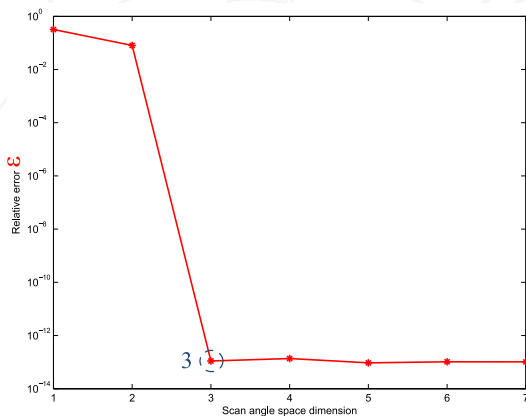
Mesh and material characteristics are extracted from the previous HFSS project

- ▶ 358' 648 DoFs
- ▶ 1100 $\tilde{\mathbf{E}}, \tilde{\mathbf{H}}$ sampling points on Γ_R , $\lambda_0/10$ resolution
- ▶ 25×25 2D-DFT coefficients
- ▶ Memory required by the full radiation model: ≈ 625 MB
- ▶ Computation timings: $\begin{cases} \text{Matrices assembly: } \approx 20 \text{ min} \\ \text{Input system solution: } \approx 25 \text{ min} \end{cases}$
- ▶ Parameterization $\Rightarrow$ pattern computation on $\phi = 0^\circ$ e $\phi = 90^\circ$ cut demands **150 ms** (233 times faster than HFSS)

Model reduction for beam steering in the XZ plane

► α scan directions

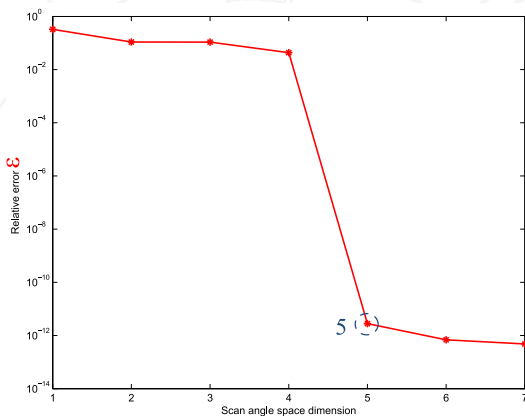
$(\theta_{s_i} \in \{0^\circ, 90^\circ, 45^\circ, 22.5^\circ, 67.5^\circ, \dots\}, \phi_{s_i} = 0^\circ)$



Model reduction for beam steering in the YZ plane

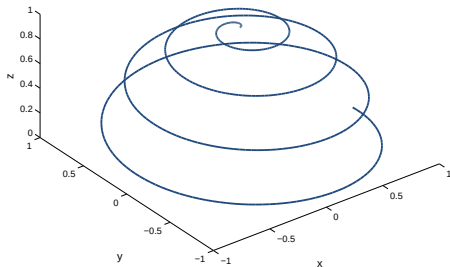
► α scan directions

$(\theta_{s_i} \in \{0^\circ, 90^\circ, 45^\circ, 22.5^\circ, 67.5^\circ, \dots\}, \phi_{s_i} = 90^\circ)$

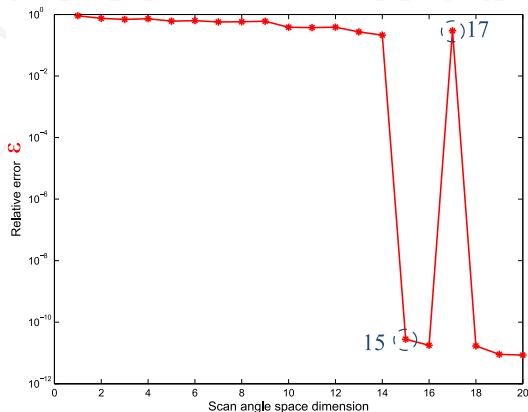


Model reduction for beam steering in the whole space

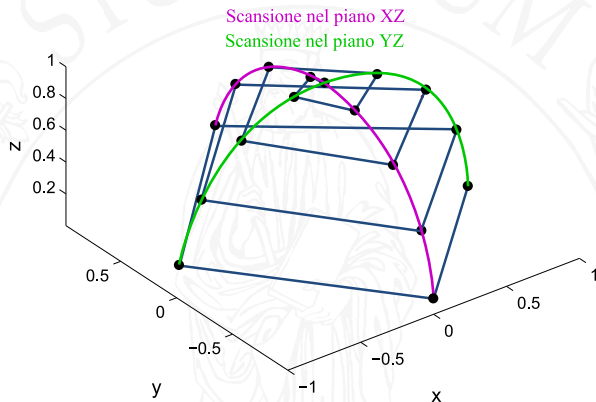
- α scan directions $(\theta_{s_i}, \phi_{s_i})$ individuated by equally spaced points on a spiraling helicoidal trajectory for the northern hemisphere



Model reduction for beam steering in the whole space



Model reduction for beam steering in the whole space



Model reduction for beam steering in the whole space

- ▶ $\alpha = 15$ (n° of antennas)
 - ▶ N° of DoFs: 358'648 $\rightarrow$ 15
 - ▶ Memory usage: 625 MB $\rightarrow$ 1.5 MB
 - ▶ Model order reduction error: $\varepsilon \leq 10^{-11}$
 - ▶ 2D-DFT truncation error: $\varepsilon_{n2f} \leq 10^{-4}$
 $\rightarrow \approx 0.01$ dB maximum pattern error

$$\Delta\theta = 0.12^\circ, \Delta\phi = 3^\circ$$

Frame computation time with the given resolution:

- ▶ for HFSS: 7 min 10 s
- ▶ for the full parameterized model: 520 ms
- ▶ for the reduced parameterized model: 63 ms

Whole movie:

- ⇒
- ▶ 10 h 20 min
 - ▶ ≈ 1 min 3 s
 - ▶ ≈ 8 s

Conclusions

- ▶ Promising technique for the analysis of large phased arrays
- ▶ Further parameterization in the frequency (k_0) and material characteristics ($\bar{\epsilon}_r, \bar{\mu}_r$) $\Rightarrow$ Working bandwidth and fabrication tolerances
- ▶ Optimization of pattern shape relying on accurately computed patterns